

Login	
Attributes	Detailed Specification
Use Case Name	Login
Actor	Manager, Staff, Customer
Purpose	This use case allows an actor to access the TechStore system using a registered account.
Pre-Conditions	The actor already has a valid account. The system is running and connected to the database.
Post-Conditions	The actor successfully logs into the system. The system redirects the actor to the appropriate interface based on their role.
Main Flow	1. The actor opens the TechStore system. 2. The system displays the Login screen. 3. The actor enters email and password and clicks the Login button. 4. The system validates the login information, identifies the actor role, records the successful login activity, and redirects the actor to the corresponding interface.
Alternative Flow	None.
Exception Flow	4a. Invalid email or password 1. At step 4, the system detects incorrect login information, displays the message: "Incorrect email or password. Please try again". 2. Return to step 3 of the Main Flow. 4b. Empty email or password 1. At step 4, the system detects that email or password is empty and displays the message: "Email and password are required". 2. Return to step 3 of the Main Flow. 4c. Account is blocked or deleted 1. At step 4, the system detects that the account status is BLOCKED or DELETED and displays the message: "Your account is blocked or no longer active". 4d. Database or system connection error 1. At step 4, the system cannot connect to the database or server and displays the message: "Unable to connect to the system. Please try again later".

Search Products

Attributes	Detailed Specification
Use Case Name	Search Products
Actor	Customer, Manager, Staff
Purpose	This use case allows an actor to search, filter, sort, and browse products in the TechStore catalog.
Pre-Conditions	The actor has successfully logged into the system. The system is running and connected to the database. Product data is available in the system.
Post-Conditions	The system displays a list of products matching the actor's search criteria.
Main Flow	1. The actor clicks on the search bar displayed on the interface. 2. The actor enters a keyword or selects a product category. 3. The system retrieves matching products from the database and displays results with basic information: product name, price, brand, category, image, and stock status. 4. The actor reviews the displayed product list.
Alternative Flow	2a. Actor applies advanced filters 1. At step 2, the actor selects filters: price range, brand, category, stock availability, promotion status, or sorting option. 2. The system updates the product list and continues at step 4 of the Main Flow. 4a. Actor navigates to another page 1. At step 4, the actor selects a page number or navigation button. 2. The system loads the next set of products and continues at step 4 of the Main Flow. 4b. Actor clears all filters 1. At step 4, the actor clicks "Clear Filters". 2. The system resets all filters and displays the default product list and continues at step 2 of the Main Flow. 4c. Actor views product details 1. At step 4, the actor selects a product from the list. 2. The system displays the detailed information of the selected product. 3. The actor returns to the product list and continues at step 4 of the Main Flow.

Exception Flow	4a. No products match the search criteria 1. At step 4, the system finds no products matching the entered keyword or selected filters, displays the message: “No products found matching your search criteria”, and provides an option to clear all filters.
	4b. Invalid filter input 1. At step 4, the system detects invalid filter input, such as an invalid price range and displays the message: “Invalid filter input. Please check your search criteria”. 2. The Actor returns to step 3 of the Main Flow.
	4c. System fails to retrieve product data 1. At step 4, the system cannot retrieve product data from the database and displays the message: “Unable to load products. Please try again later”.

View Product Details	
Attributes	Detailed Specification
Use Case Name	View Product Details
Actor	Customer, Manager, Staff
Purpose	This use case allows an actor to view detailed information about a selected product.
Pre-Conditions	The actor has successfully logged into the system. The selected product exists in the system.
Post-Conditions	The product details are displayed successfully.
Main Flow	1. The actor selects a product from the product search result. 2. The system retrieves detailed information of the selected product and displays product details, including product name, description, price, images, brand, category, stock status, and available bundle services. 3. The actor reviews the product details.
Alternative Flow	3a. Actor returns to product list 1. At step 3, the actor selects the Back to Product List option. 2. The system returns to the product search result.
Exception Flow	2a. Product no longer exists 1. At step 2, the system detects that the selected product no longer exists, displays the message: “This product is no longer available”, and returns to the product search result. 2b. System fails to load product details 1. At step 2, a database or system error occurs. 2. The system displays the message: “Unable to load product details. Please try again later”.

Manage Cart	
Attributes	Detailed Specification
Use Case Name	Manage Cart
Actor	Customer
Purpose	This use case allows the Actor to add products to the shopping cart, manage cart items, and place an order through payment.
Pre-Conditions	The Actor has successfully logged into the system. The required data is available in the system. At least one product is available for purchase.
Post-Conditions	The Actor's cart is updated successfully. Selected cart items are used to create a new order after successful checkout. Payment information is processed and recorded. Inventory quantity is updated for purchased products. All cart items are removed from the cart after successful checkout. The Actor receives the order confirmation and payment result.
Main Flow	1. The Actor selects a product from the product list or product detail page and clicks the Add to Cart button. 2. The system checks the selected product's availability, adds the selected product to the Actor's cart, and updates the cart item count. 3. The Actor opens the Cart screen. 4. The system displays all cart items, including product name, unit price, quantity, selected bundle services, item subtotal, and cart total. 5. The Actor reviews the cart items and clicks Proceed to Checkout. 6. The system validates the cart items and available stock, and Payment is included. 7. The system receives the payment result from Payment, creates a new order from all cart items, saves order information, order item details, selected bundle services, payment information, and payment status, updates inventory quantity, removes all cart items from the Actor's cart, and displays the order confirmation, payment result, and final payment amount.

Alternative Flow	2a. Product already exists in the cart 1. At step 2, the system detects that the selected product already exists in the cart, increases the quantity of the existing cart item, updates the cart item count, cart total and continues at step 3 of the Main Flow. 5a. Actor changes item quantity 1. At step 5, the Actor changes the quantity of a cart item. 2. The system validates the updated quantity, updates the cart item quantity, recalculates the item subtotal, cart total and continues at step 5 of the Main Flow. 5b. Actor removes a cart item 1. At step 5, the Actor removes a cart item from the cart. 2. The system removes the item, recalculates the cart total and continues at step 4 of the Main Flow. 5c. Actor adds bundle services 1. At step 5, the Actor selects a bundle service option for a cart item. 7a. Actor views invoice 1. At step 7, the Actor selects the View Invoice option. 2. The system navigates the Actor to the invoice detail page.
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Exception Flow	2a. Product is out of stock 1. At step 2, the system detects that the selected product is out of stock and displays the message: "This product is currently out of stock." 2. The product is not added to the cart. 5a. Invalid quantity 1. At step 5, the Actor enters an invalid quantity. 2. The system rejects the quantity update, displays the message: "Invalid quantity. Please enter a valid quantity." and returns to step 5 of the Main Flow. 5b. Cart is empty 1. At step 5, the Actor clicks Proceed to Checkout while the cart has no items. 2. The system displays the message: "Your cart is empty. Please add products before checkout." and returns to step 1 of the Main Flow. 6a. Cart item becomes unavailable before checkout 1. At step 6, the system detects that one or more cart items are no longer available in stock, highlights the unavailable items, displays the message: "Some items in your cart are no longer available. Please remove or update them to continue." and returns to step 5 of the Main Flow. 7a. Payment fails 1. At step 7, the system receives a failed, cancelled, or expired payment result, does not create the order, displays the message: "Payment failed. Please try again or choose another payment method." and returns to step 5 of the Main Flow. 7b. System fails to save order or payment information 1. At step 7, a database or system error occurs while saving order or payment information. 2. The system does not confirm the order and displays the message: "Unable to complete your order. Please try again later." 7c. System fails to update inventory quantity 1. At step 7, the system cannot update inventory quantity due to a database or system error, marks the order as pending review, and displays the message: "Unable to update inventory information. Please contact support or try again later."
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Add Bundle Services	
Attributes	Detailed Specification
Use Case Name	Add Bundle Services
Actor	Customer
Purpose	This use case allows the Actor to add optional bundle services, such as warranty extension or screen protection, to a selected cart item.
Pre-Conditions	The Actor has successfully logged into the system. The Actor has at least one item in the cart. At least one active bundle service is available in the system.
Post-Conditions	The selected bundle service is added to the cart item. The cart item subtotal is updated. The checkout total is recalculated.
Main Flow	1. The Actor selects a cart item. 2. The system displays available bundle services for the selected cart item. 3. The Actor selects one or more bundle services. 4. The system validates the selected bundle services, adds the selected bundle services to the cart item, recalculates the cart item subtotal and checkout total, and displays the updated cart information.
Alternative Flow	3a. Actor removes a selected bundle service 1. At step 3, the Actor deselects a previously selected bundle service. 2. The system removes the bundle service from the cart item, recalculates the cart item subtotal and checkout total and continues at step 4 of the Main Flow. 3b. Actor cancels adding bundle services 1. At step 3, the Actor cancels the bundle service selection. 2. The system keeps the cart item unchanged.
Exception Flow	4a. Bundle service is inactive 1. At step 4, the system detects that the selected bundle service is inactive and displays the message: "This bundle service is currently unavailable". and returns to step 2 of the Main Flow. 4b. System fails to update cart item 1. At step 4, a database or system error occurs. 2. The system displays the message: "Unable to add bundle service. Please try again later".

Payment

Attributes	Detailed Specification
Use Case Name	Payment
Actor	Customer
Purpose	This use case allows the Actor to select a payment method, and complete payment for cart items during checkout.
Pre-Conditions	The Actor has successfully logged into the system. The Actor has selected at least one cart item for checkout. The checkout total has been calculated. The system is connected to the payment service or payment method handler.
Post-Conditions	Payment information is recorded by the system. A successful online payment is recorded with transaction information. A COD payment is recorded as pending. A failed online payment attempt is recorded without creating an order.
Main Flow	1. The system displays the payment screen including cart items, cart total, and available payment methods: MoMo, VNPay, and COD. 2. The Actor reviews the cart items and cart total and selects a payment method 3. The system displays the payment summary including cart items, final amount, and selected payment method. 4. The Actor confirms the payment. 5. The system processes the payment according to the selected payment method, receives the payment result, records the payment result.
Alternative Flow	3a. Actor selects COD 1. At step 3, the Actor selects COD as the payment method. 2. The system records the payment method as COD, sets the payment status to Awaiting Confirmation and continues at step 4 of the Main Flow. 3b. Actor changes payment method 1. At step 3, the Actor selects another available payment method. 2. The system updates the payment summary and continues at step 4 of the Main Flow. 5a. Actor cancels payment 1. At step 5, the Actor cancels the payment confirmation. 2. The system does not process the payment and returns the Actor to the payment screen. 6a. Actor views invoice 1. At step 6, after the order is created successfully, the Actor selects the View Invoice option. 2. The system navigates the Actor to the invoice detail page.

Exception Flow	6a. Online payment fails 1. At step 6, the payment provider returns a failed payment result. 2. The system records the failed payment attempt, displays the message: "Payment failed. Please try again or choose another payment method." and returns to step 1 of the Main Flow. 6b. Payment is cancelled 1. At step 6, the payment provider returns a cancelled payment result. 2. The system records the payment attempt as Cancelled, displays the message: "Payment was cancelled. Please try again." and returns to step 1 of the Main Flow. 6c. Payment service is unavailable 1. At step 6, the system cannot connect to the payment provider and displays the message: "Payment service is currently unavailable. Please try again later." 2. The payment process terminates unsuccessfully.

View Invoice	
Attributes	Detailed Specification
Use Case Name	View Invoice
Actor	Customer
Purpose	This use case allows the Actor to view and export invoice information for a completed or valid order.
Pre-Conditions	The Actor has successfully logged into the system. The selected order exists in the system. The invoice information for the selected order is available.
Post-Conditions	The invoice is displayed successfully. The Actor can export the invoice as a PDF file.
Main Flow	1. The Actor selects the View Invoice option from the order confirmation screen. 2. The system creates invoice information for the selected order and displays invoice details, including invoice ID, order ID, product list, quantities, unit prices, discounts, VAT, payment method, final amount, and issued date. 3. The Actor reviews the invoice.

Alternative Flow	3a. Actor exports invoice as PDF - At step 3, the Actor clicks Export Invoice PDF. - The system generates the invoice PDF, downloads the invoice PDF to the Actor's device and continues at step 3 of the Main Flow. 3b. Actor returns to order information - At step 3, the Actor selects Back to Order. - The system returns to the previous order information screen.
Exception Flow	3c. System fails to generate invoice PDF - At step 2, the Actor requests PDF export and the system fails to generate the PDF. - The system displays the message: "Unable to generate invoice PDF. Please try again later" and returns to step 3 of the Main Flow.

Subscribe to Notifications	
Attributes	Detailed Specification
Use Case Name	Subscribe to Notifications
Actor	Customer
Purpose	This use case allows the Actor to subscribe to automated notifications for a selected product, such as stock changes, price changes, or promotional updates.
Pre-Conditions	The Actor has successfully logged into the system. The selected product exists in the system.
Post-Conditions	The Actor's subscription status for the selected product is updated. The system stores the Actor's product subscription information. The notification system can identify the Actor when relevant product changes occur.
Main Flow	- The Actor opens the product detail page. - The system displays the product details and a heart button indicating the current subscription status. - The Actor clicks the heart button. - The system saves the subscription for the selected product, updates the heart button to the subscribed state, and displays the message: "You have successfully subscribed to product updates."
Alternative Flow	3a. Actor unsubscribes from a product At step 3, the Actor clicks the heart button while already subscribed.

	The system removes the subscription, updates the heart button to the unsubscribed state, and displays the message: "You have successfully unsubscribed from product updates."
Exception Flow	4a. Product is no longer available - At step 4, the system detects that the selected product no longer exists and displays the message: "This product is no longer available for subscription." 4b. System fails to save subscription - At step 4, a database or system error occurs and the system displays the message: "Unable to update subscription. Please try again later."

View Membership Tier	
Attributes	Detailed Specification
Use Case Name	View Membership Tier
Actor	Customer
Purpose	This use case allows the Actor to view their current membership tier, accumulated spending, and membership benefits.
Pre-Conditions	The Actor has successfully logged into the system. Membership tier configuration exists in the system.
Post-Conditions	The Actor's current membership tier is displayed. The Actor's total accumulated spending is displayed. The benefits associated with the tier are displayed.
Main Flow	- The Actor navigates to the Personal Profile page and selects the Membership Tier section. - The system retrieves the Actor's total accumulated spending, determines the Actor's current membership tier based on predefined thresholds, and displays the Actor's current tier, total accumulated spending, and active benefits.
Alternative Flow	2a. Actor's tier is upgraded after completed order - After a new order is marked as Completed, the system adds the order value to the Actor's total accumulated spending. - The system recalculates the Actor's membership tier, updates the Actor's membership tier, and displays a congratulatory notification to the Actor.

Exception Flow	2a. Actor account is restricted - At step 2, the system detects that the Actor's account is restricted, does not display the Membership Tier section, and displays the message: "Your account has been restricted. You are unable to access membership benefits". 2b. System fails to retrieve membership information - At step 2, a database or system error occurs. - The system displays the message: "Unable to load membership information. Please try again later".
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Track Order History	
Attributes	Detailed Specification
Use Case Name	Track Order History
Actor	Customer
Purpose	This use case allows the Actor to view purchase history, track order status, and access invoice information.
Pre-Conditions	The Actor has successfully logged into the system. The Actor has placed at least one order.
Post-Conditions	The Actor can view order history. The Actor can view details and current status of selected orders. The Actor can access invoice information for an order.
Main Flow	- The Actor navigates to the Personal Profile page and selects the Purchase History tab. - The system retrieves all orders associated with the Actor's account and displays the orders sorted from most recent to oldest, including order ID, order date, order status, and total amount. - The Actor selects an order from the list. - The system displays the order details including order ID, order date, payment method, product list, quantities, unit prices, discounts, VAT, and final payment amount.
Alternative Flow	2a. Actor filters or sorts orders - At step 2, the Actor selects a status filter or enters a custom date range. - The system updates the order list based on the selected criteria and continues at step 3 of the Main Flow. 4a. Actor views invoice - At step 4, the Actor selects the View Invoice option. - The system navigates the Actor to the invoice detail page.

Exception Flow	2a. Actor has no order history - At step 2, the system finds no orders associated with the Actor's account and displays the message: "You have not placed any orders yet." 2b. Actor session has expired - At step 2, the system detects that the Actor's login session has expired, displays the message: "Your session has expired. Please log in again", and redirects the Actor to the Login screen. 2c. System fails to retrieve order history - At step 2, a database or system error occurs and the system displays the message: "Unable to load order history. Please try again later."
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Import Products	
Attributes	Detailed Specification
Use Case Name	Import Products
Actor	Staff, Manager
Purpose	This use case allows Actor to import products into warehouse and update inventory quantity.
Pre-Conditions	The Actor has successfully logged into the system. The required data is available in the system.
Post-Conditions	Product import information is saved successfully. Inventory quantity is updated. The import action is recorded.
Main Flow	- The Actor opens the Product Management screen. - The system displays the product management interface. - The Actor selects the Import Products button. - The system displays the import product form. - The Actor enters product import information and clicks the Import button. - The system validates the entered information. - The Actor confirms the import request. - The system stores the import information, updates inventory quantity, records the import action in ImportLog, and displays a success notification.

Alternative Flow	5a. Actor imports an existing product 1. At step 5, the Actor selects an existing product. 2. The system loads existing product information. 3. The Actor enters import quantity and import price and continues at step 5 of the Main Flow. 5b. Actor imports a new product 1. At step 5, the Actor enters new product information. 2. The system prepares a new product record and continues at step 5 of the Main Flow.
Exception Flow	6a. Invalid or missing product information 1. At step 6, the system detects invalid or missing input fields, displays an error message requesting valid information and returns to step 5 of the Main Flow. 6b. Duplicate product information 1. At step 6, the system detects duplicate product information for a new product, displays the message: "Product information already exists" and returns to step 5 of the Main Flow. 8a. Database or system error 1. At step 8, the system cannot save import information and displays the message: "Unable to import products. Please try again later".

Export Products	
Attributes	Detailed Specification
Use Case Name	Export Products
Actor	Staff, Manager
Purpose	This use case allows Actor to export products and update inventory quantity.
Pre-Conditions	The Actor has successfully logged into the system. The required data is available in the system. Products are available in inventory.
Post-Conditions	Product export information is saved successfully. Inventory quantity is updated. A receipt is generated successfully. Inventory change status is displayed. The export action is recorded.

Main Flow	1. The Actor opens the Product Management screen. 2. The system displays the product management interface. 3. The Actor selects the Export Products button. 4. The system displays the export product form. 5. The Actor enters product export information and clicks the Export button. 6. The system validates the entered information and checks product availability in inventory. 7. The Actor confirms the export request. 8. The system saves the export information, updates inventory quantity, records the export action in ExportLog, Generate Receipt and Notify Inventory Change Status is included. 9. The system displays a success message.
Alternative Flow	None.
Exception Flow	6a. Invalid or missing export product information 1. At step 6, the system detects invalid or missing input fields, displays an error message requesting valid information and returns to step 5 of the Main Flow. 6b. Insufficient product quantity in inventory 1. At step 6, the system detects that the requested export quantity exceeds available stock, displays the message: "Insufficient product quantity in inventory" and returns to step 5 of the Main Flow. 8a. Database or system error 1. At step 8, the system cannot save export information and displays the message: "Unable to export products. Please try again later". 8b. Receipt generation failsh 1. At step 8, the system displays the message: "Products were exported, but the receipt could not be generated" and records the issue for later review. 9a. Inventory notification fails 1. At step 9, the system returns a notification failure. 2. The system displays the message: "Inventory was updated, but notification status could not be displayed" and records the issue for later review.

Generate Receipt	
Attributes	Detailed Specification
Use Case Name	Generate Receipt

Actor	Staff
Purpose	This use case allows the system to generate a receipt after product export information has been saved successfully.
Pre-Conditions	The Actor has successfully logged into the system. Product export information has been validated. ExportLog information is available.
Post-Conditions	A receipt is generated successfully. The receipt is linked to the export transaction. The receipt can be viewed, printed, or downloaded.
Main Flow	1. The system receives a receipt generation request from Export Products, retrieves export information from ExportLog, calculates the receipt details, generates the receipt, links the receipt to the export record, and returns the generated receipt result to Export Products.
Alternative Flow	1a. Actor downloads the receipt 1. After step 1, the Actor selects the Download Receipt option. 2. The system generates a downloadable receipt file and provides the file to the Staff. 1b. Actor prints the receipt 1. After step 1, the Actor selects the Print Receipt option. 2. The system sends the receipt to the print function.
Exception Flow	1a. Export information is not found 1. At step 1, the system cannot find the export information and displays the message: "Export information was not found". 1b. Receipt generation fails 1. At step 1, the system cannot generate the receipt and displays the message: "Unable to generate receipt. Please try again later".

Notify Inventory Change Status	
Attributes	Detailed Specification
Use Case Name	Notify Inventory Change Status
Actor	Staff, Manager
Purpose	This use case allows the system to display inventory update status after product export and notify relevant users when inventory status changes.
Pre-Conditions	The Actor has successfully logged into the system. Product export information has been saved. Inventory quantity has been updated.
Post-Conditions	Inventory change status is displayed to the Actor. Relevant inventory status information is recorded.

Main Flow	1. The system receives an inventory change notification request from Export Products, retrieves the updated product inventory quantity, determines the product inventory status, records the inventory change status, displays the inventory update confirmation to the Actor, and returns the notification result to Export Products.
Alternative Flow	1a. Product becomes out of stock 1. At step 1, the system determines that the product quantity reaches zero, sets the product inventory status to Out of Stock, prepares notification information for subscribed Customers and continues at step 1 of the Main Flow. 1b. Product remains available 1. At step 1, the system determines that the product quantity remains greater than zero, sets the product inventory status to Available and continues at step 1 of the Main Flow.
Exception Flow	1a. Inventory data cannot be retrieved 1. At step 1, the system cannot retrieve updated inventory data and displays the message: "Unable to retrieve inventory status". 1b. System fails to record inventory change status 1. At step 1, a database or system error occurs. 2. The system displays the message: "Unable to record inventory change status".

View Warehouse Log	
Attributes	Detailed Specification
Use Case Name	View Warehouse Log
Actor	Manager
Purpose	This use case allows the Actor to view warehouse log records based on product import logs and product export logs.
Pre-Conditions	The Actor has successfully logged into the system. Warehouse log data is available. The required data is available in the system.
Post-Conditions	The system displays warehouse log records based on ImportLog and ExportLog data. The Actor can view detailed information of each import or export record.

Main Flow	1. The Actor opens the Warehouse Management screen. 2. The system displays the warehouse log section. 3. The Actor selects View Warehouse Log. 4. The system retrieves product import records from ImportLog and product export records from ExportLog and displays the warehouse log list, including log type, product name, quantity, performed by, status, and created time. 5. The Actor enters or selects filter conditions. 6. The system validates the filter conditions and displays matching warehouse log records. 7. The Actor selects a warehouse log record. 8. The system displays the detailed information of the selected warehouse log record.
Alternative Flow	5a. Actor views only import logs 1. At step 5, the Actor selects the Import Log filter. 2. The system displays only product import records from ImportLog and continues at step 7 of the Main Flow. 5b. Actor views only export logs 1. At step 5, the Actor selects the Export Log filter. 2. The system displays only product export records from ExportLog and continues at step 7 of the Main Flow. 5c. Actor exports warehouse log list 1. At step 5, the Actor selects the Export function and chooses a file format, such as CSV or Excel. 2. The system generates an export file based on the current filter conditions and provides the generated file for download. 8a. Selected record is ImportLog 1. At step 8, the system displays import quantity, import price, import date, status, note, product information, and Actor who performed the import. 8b. Selected record is ExportLog 1. At step 8, the system displays export quantity, export date, reason, status, product information, and Actor who performed the export.

Exception Flow	6a. Invalid filter input 1. At step 6, the system detects invalid filter values, such as an incorrect date range displays the message: “Invalid filter input. Please check the selected conditions” and returns to step 5 of the Main Flow.
	6b. No warehouse log records found 1. At step 6, the system finds no ImportLog or ExportLog records matching the selected conditions and displays the message: “No warehouse log records were found matching the selected criteria” and returns to step 5 of the Main Flow.
	4a. System fails to load warehouse log data 1. At step 4, a database or system error occurs while retrieving ImportLog or ExportLog data. 2. The system displays the message: “Unable to load warehouse log data. Please try again later”.

View Login Log	
Attributes	Detailed Specification
Use Case Name	View Login Log
Actor	Manager
Purpose	This use case allows the Actor to view login activity records in the TechStore system.
Pre-Conditions	The Actor has successfully logged into the system. The required data is available in the system.
Post-Conditions	Login log information is displayed successfully. The Actor can review login activity records.
Main Flow	1. The Actor opens the System Management screen. 2. The system displays the system management interface. 3. The Actor selects View Login Log. 4. The system retrieves login log information from the database and displays the login log list, including login time, email, role, login status, and when available. 5. The Actor reviews the login log list.

Alternative Flow	5a. Actor filters login logs 1. At step 5, the Actor enters filter conditions such as email, role, login status, or date range. 2. The system validates the filter conditions, displays matching login log records and continues at step 5 of the Main Flow. 5b. Actor exports login logs 1. At step 5, the Actor selects the Export function. 2. The system generates a login log file and provides the file for download.
Exception Flow	4a. Unable to retrieve login log information 1. At step 4, the system cannot retrieve login log data from the database and displays the message: "Unable to load login log information. Please try again later". 2a. Actor does not have permission 1. At step 2, the system detects that the current account does not have permission to view login logs and displays the message: "You do not have permission to view login logs".

View Payment Log	
Attributes	Detailed Specification
Use Case Name	View Payment Log
Actor	Manager
Purpose	This use case allows the Actor to view payment transaction logs, including successful, failed, pending, cancelled, and refunded payments.
Pre-Conditions	The Actor has successfully logged into the system. Payment log data is available. The required data is available in the system.
Post-Conditions	The payment log list is displayed. The Actor can review transaction information.
Main Flow	1. The Actor navigates to the View Payment Log section from the Dashboard. 2. The system retrieves payment transaction records from the database and displays the payment transaction list, including order ID, customer, amount, payment method, payment status, and timestamp. 3. The Actor reviews the payment log.

Alternative Flow	3a. Actor filters by date range 1. At step 3, the Actor selects a start date and end date and clicks Apply. 2. The system displays only payment transactions within the selected date range and continues at step 3 of the Main Flow. 3b. Actor filters by payment status 1. At step 3, the Actor selects a payment status such as Success, Failed, Pending, Cancelled, or Refunded. 2. The system displays payment transactions matching the selected status and continues at step 3 of the Main Flow. 3c. Actor filters by payment method 1. At step 3, the Actor selects a payment method such as MoMo, VNPay, or COD. 2. The system displays payment transactions matching the selected payment method and continues at step 3 of the Main Flow. 3d. Actor views payment transaction details 1. At step 3, the Actor selects a payment transaction. 2. The system displays detailed transaction information, including transaction ID, order ID, customer information, amount, payment method, status, paid time, failure reason, and information when available
Exception Flow	2a. System fails to retrieve payment logs 1. At step 2, a database or system error occurs. 2. The system displays the message: "Unable to load payment logs. Please try again later". 2b. No payment logs found 1. At step 2, the system finds no payment logs matching the selected criteria and displays the message: "No payment records found".

Report Revenue	
Attributes	Detailed Specification
Use Case Name	Report Revenue
Actor	Manager
Purpose	This use case allows the Actor to view revenue reports within a selected time period, including total revenue, total orders, best-selling products, revenue by category, revenue by brand, and payment method breakdown.

Pre-Conditions	The Actor has successfully logged into the system. At least one completed order exists in the system.
Post-Conditions	The revenue report is displayed based on the selected filters. The Actor can review revenue statistics.
Main Flow	1. The Actor navigates to the Report Revenue module from the Dashboard. 2. The system displays the default revenue report for the current month and displays total revenue, total orders, and revenue trend chart. 3. The Actor selects a time range, such as Daily, Weekly, Monthly, or Custom and clicks Apply. 4. The system retrieves completed orders within the selected time range, recalculates revenue statistics, and displays total revenue, revenue by product category, revenue by brand, top 5 best-selling products, and revenue by payment method. 5. The Actor reviews the report.
Alternative Flow	3a. Actor filters by category 1. At step 3, the Actor selects a product category. 2. The system displays revenue statistics for the selected category and continues at step 5 of the Main Flow. 3b. Actor filters by brand 1. At step 3, the Actor selects a product brand. 2. The system displays revenue statistics for the selected brand and continues at step 5 of the Main Flow. 3c. Actor filters by payment method 1. At step 3, the Actor selects a payment method such as MoMo, VNPay, or COD. 2. The system displays revenue collected through the selected payment method and continues at step 5 of the Main Flow. 5a. Actor exports revenue report 1. At step 5, the Actor selects the Export Report option. 2. The system generates the report file and provides the file for download.
Exception Flow	4a. No completed orders found in the selected period 1. At step 4, the system finds no completed orders in the selected range, displays the message: "No revenue data found for the selected period", and shows zero for all figures, then returns to step 3 of the Main Flow. 4b. System fails to load report data

	At step 4, a connection or database error occurs and the system displays the message: "Unable to load report data. Please try again later."
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Recover and Restore Data	
Attributes	Detailed Specification
Use Case Name	Recover and Restore Data
Actor	Manager
Purpose	This use case allows the Actor to recover and restore application data after data loss, corruption, accidental deletion, or system failure by restoring from a verified backup.
Pre-Conditions	The Actor has successfully logged into the system. The Actor is authorized to perform restore operations. Backups exist in the configured backup storage location. Backup metadata, including timestamp, type, and checksum, is available.
Post-Conditions	The target data store is restored to the selected recovery point. The restored data passes validation and integrity checks. The system returns to normal operation. The restore action is recorded in the audit log.
Main Flow	- The Actor opens the Backup and Restore section from the Admin Dashboard. - The system displays available recovery points. - The Actor selects a recovery point, chooses Full Restore as the restore scope, and confirms the restore operation. - The system displays selected backup information and downtime warning, places the application into maintenance mode, verifies the selected backup, creates a temporary recovery snapshot, restores data to the selected recovery point, performs validation and integrity checks, returns the application to normal operation, records the restore activity in the audit log, and notifies the Actor that the restore operation completed successfully.
Alternative Flow	3a. Actor performs a partial restore - At step 3, the Actor selects Partial Restore and chooses the specific data or module to restore. - The system validates the selected restore scope, restores only the selected data, and continues at step 4 of the Main Flow. 3b. Actor cancels restore operation - At step 3, the Actor cancels the confirmation.

	The system does not start the restore operation and keeps the current data unchanged.
Exception Flow	4a. Selected backup is missing or invalid - At step 4, the system detects that the selected backup is missing, incomplete, or invalid, stops the restore process, displays the message: "The selected backup file is missing or corrupted. Please choose another recovery point", and recommends another available recovery point when possible. 4b. Restore process fails during execution - At step 4, the restore process fails due to system or storage failure, the system keeps the application in maintenance mode, restores the previous stable state using the temporary recovery snapshot, displays the message: "Restore process failed due to a system error. The previous stable state has been recovered", and records the failure in the audit log. 4c. Restored data is incompatible - At step 4, the system detects that restored data is incompatible with the current application version, keeps the application in maintenance mode, displays the message: "The restored backup is incompatible with the current application version", and records the incident in the audit log.

Block Account	
Attributes	Detailed Specification
Use Case Name	Block Account
Actor	Manager
Purpose	This use case allows the Actor to block a selected account
Pre-Conditions	The Actor has successfully logged into the system. The selected account exists in the system.
Post-Conditions	The selected account status is changed to BLOCKED. The blocked account is prevented from logging into the system. The account's active login session is removed.

Main Flow	1. The Actor opens the Account Management screen. 2. The system displays the account list. 3. The Actor selects an account. 4. The system displays the selected account information. 5. The Actor clicks the Block Account button. 6. The system displays a confirmation message. 7. The Actor confirms the blocking action. 8. The system checks the account status, changes the account status to BLOCKED, removes the account's active login session, records the blocking action, and displays a success message.
Alternative Flow	7a. Actor cancels blocking account 1. At step 7, the Actor cancels the confirmation. 2. The system keeps the account status unchanged.
Exception Flow	8a. Account already has BLOCKED status 1. At step 8, the system detects that the selected account already has BLOCKED status, displays the message: "This account already has Blocked status", and does not change any account data. 8b. System rules do not allow blocking this account 1. At step 8, the system detects that system rules do not allow blocking the selected account, displays the message: "System rules do not allow blocking this account", and does not change any account data. 8c. System fails to block account 1. At step 8, a system error occurs while updating the account status or removing the active login session. 2. The system displays the message: "Unable to block account. Please try again later" and does not change any account data.

Add Staff	
Attributes	Detailed Specification
Use Case Name	Add Staff
Actor	Manager
Purpose	This use case allows the Actor to add a new Staff and create a account for that Staff.
Pre-Conditions	The Actor has successfully logged into the system. The required data is available in the system.
Post-Conditions	A new Staff profile is created and saved in the system. A new Staff account is created in the system.

Main Flow	1. The Actor opens the Staff Management function and selects Add Staff. 2. The system displays the Staff creation form. 3. The Actor enters Staff information, including full name, email, phone number, staff code, hire date, initial password and submits the form. 4. The system validates the entered Staff information. 5. Create Account use case is included. 6. The system creates a new Staff profile and links it to the created account, saves the new Staff information, records the Staff creation action, and displays a success message.
Alternative Flow	3a. Actor cancels adding Staff 1. At step 3, the Actor cancels the Staff creation form. 2. The system discards the entered information and does not create Staff data or account data.
Exception Flow	4a. Required Staff information is missing 1. At step 4, the system detects missing required information, displays the message: "Staff information is missing", does not create the Staff profile, and returns to step 3 of the Main Flow. 4b. Invalid Staff information 1. At step 4, the system detects invalid Staff information such as invalid email format or phone number format, displays the message: "Invalid Staff information", does not create the Staff profile, and returns to step 3 of the Main Flow. 4c. Duplicate Staff email detected 1. At step 4, the system detects that the email already belongs to another account, displays the message: "Email already exists", does not create the Staff profile, and returns to step 3 of the Main Flow. 5a. Account creation fails 1. At step 5, Create Account use case returns an account creation failure, the system does not save Staff information, displays the message: "Unable to create Staff account. Please try again later", and the use case terminates unsuccessfully. 6a. System fails to save Staff information 1. At step 6, a system error occurs while saving Staff information and the system displays the message: "Unable to add Staff. Please try again later" and does not save any Staff data.

Create Account	
Attributes	Detailed Specification
Use Case Name	Create Account
Actor	Manager
Purpose	This use case allows Actor to create a new account for a Staff member during the Add Staff process.
Pre-Conditions	The Actor has successfully logged into the system. Account information is provided from Add Staff use case. The provided email does not belong to an existing active account.
Post-Conditions	A new account is created successfully. The account is assigned to the Staff role. The account status is set to ACTIVE. The account creation action is recorded by the system.
Main Flow	1. The system receives account creation information from Add Staff use case, validates email, initial password, and account role, encrypts the initial password, creates a new account, sets the account status to ACTIVE, assigns the Staff role to the account, records the account creation action, and returns the created account result to Add Staff use case.
Alternative Flow	1a. System requires password reset on first login 1. At step 1, the system applies first-login password reset policy, marks the account as requiring password reset and continues at step 1 of the Main Flow.
Exception Flow	1a. Invalid account information 1. At step 1, the system detects invalid email or password format and returns an account creation failure to Add Staff use case. 1b. Duplicate account email 1. At step 1, the system detects that the email already exists and returns a duplicate email error to Add Staff use case. 1c. System fails to create account 1. At step 1, a database or system error occurs. 2. The system returns an account creation failure to Add Staff use case.

Delete Staff	
Attributes	Detailed Specification
Use Case Name	Delete Staff
Actor	Manager
Purpose	This use case allows the Actor to remove a Staff member from the active Staff list and remove the Staff account access from the TechStore system.

Pre-Conditions	The Actor has successfully logged into the system. The Staff to be deleted exists in the system. The Actor has permission to manage Staff.
Post-Conditions	The selected Staff is removed from the active Staff list. The Staff account access is removed from the system. The Staff account can no longer access the system.
Main Flow	1. The Actor opens the Staff Management function. 2. The system displays the Staff list. 3. The Actor searches or filters Staff and selects a Staff member. 4. The system displays the selected Staff details. 5. The Actor selects Delete Staff. 6. The system displays a deletion confirmation message. 7. The Actor confirms the deletion. 8. Delete Account is included. 9. The system removes the Staff from the active Staff list, records the Staff deletion action, and displays a success message.
Alternative Flow	3a. Actor views all Staff 1. At step 3, the Actor leaves the search criteria empty. 2. The system keeps displaying the full Staff list and continues at step 3 of the Main Flow. 7a. Actor cancels deleting Staff 1. At step 7, the Actor cancels the deletion confirmation. 2. The system keeps the Staff record and account access unchanged and the use case ends.
Exception Flow	3a. No Staff matches the search criteria 1. At step 3, the system finds no matching Staff, displays the message: "No Staff found", and returns to step 3 of the Main Flow. 8a. Delete Account fails 1. At step 8, Delete Account use case returns an account deletion failure and the system does not remove the Staff from the active Staff list and displays the message: "Unable to remove Staff account access. Please try again later." 9a. System fails to remove Staff from active list 1. At step 9, a system error occurs while removing the Staff from the active Staff list and the system displays the message: "Unable to delete Staff. Please try again later" and does not change any Staff data.

Delete Account	
Attributes	Detailed Specification
Use Case Name	Delete Account
Actor	Manager
Purpose	This use case allows the system to remove or deactivate account access for a selected Staff member during the Delete Staff process.
Pre-Conditions	The Manager has successfully logged into the system. The selected Staff account exists in the system. The account is linked to the Staff member being deleted.
Post-Conditions	The account can no longer log into the system. Active login sessions of the account are removed. The account deletion action is recorded by the system.
Main Flow	1. The system receives an account deletion request from Delete Staff, retrieves the selected Staff account, validates the account deletion rule, removes active login sessions of the account, records the account deletion action, and returns the account deletion result to Delete Staff.
Alternative Flow	None.
Exception Flow	1a. Account does not exist 1. At step 1, the system cannot find the selected Staff account and returns an account deletion failure to Delete Staff. 2. The use case terminates unsuccessfully. 1b. System rules do not allow deleting this account 1. At step 1, the system detects that the account cannot be deleted due to system rules and returns a deletion-not-allowed result to Delete Staff. 2. The use case terminates unsuccessfully. 1c. System fails to remove active sessions 1. At step 1, the system fails to remove active login sessions and returns an account deletion failure to Delete Staff. 2. The use case terminates unsuccessfully.

Apply Promotion	
Attributes	Detailed Specification
Use Case Name	Apply Promotion
Actor	Customer
Purpose	This use case allows the Actor to apply a valid promotion code during checkout to receive a discount on the cart total.

Pre-Conditions	The Actor has successfully logged into the system. The Actor is on the Payment screen. At least one active promotion exists in the system.
Post-Conditions	The valid promotion code is applied and the discount is reflected in the final amount.
Main Flow	1. The Actor enters a promotion code in the promotion code input field and clicks Apply. 2. The system validates the promotion code against active promotions in the database. 3. The system applies the discount to the cart total, displays the discount amount and updated final amount, and updates the promotion code input field to the applied state.
Alternative Flow	3a. Actor removes applied promotion 1. At step 3, the Actor clicks the Remove button next to the applied promotion code. 2. The system removes the discount, restores the original cart total, and resets the promotion code input field to the empty state. 1a. Actor re-enters a different promotion code 1. At step 1, the Actor clears the current promotion code and enters a new one. 2. The system validates the new promotion code and applies the discount if valid, then continues at step 3 of the Main Flow.
Exception Flow	2a. Promotion code does not exist 1. At step 2, the system finds no promotion matching the entered code, displays the message: "Promotion code does not exist. Please check and try again", and the cart total remains unchanged. 2b. Promotion code is expired 1. At step 2, the system detects that the promotion code is expired, displays the message: "This promotion code has expired", and the cart total remains unchanged. 2c. Promotion code has already been used 1. At step 2, the system detects that the promotion code has already been used by the Actor, displays the message: "This promotion code has already been used", and the cart total remains unchanged. 2d. System fails to validate promotion code

	At step 2, a database or system error occurs and the system displays the message: "Unable to validate promotion code. Please try again later", and the cart total remains unchanged.
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Create Promotion	
Attributes	Detailed Specification
Use Case Name	Create Promotion
Actor	Manager
Purpose	Allows the Manager to create a new promotion with discount details and applicable conditions.
Pre-Conditions	The Manager is logged into the system.
Post-Conditions	A new promotion is saved in the system with status ACTIVE or INACTIVE based on start date.
Main Flow	- The Manager navigates to the Promotion Management screen and clicks "Create Promotion". - The system displays the Create Promotion form including: promotion name, promotion code, discount type (percentage / fixed amount), discount value, start date, end date, minimum order value, usage limit per customer, and total usage limit. - The Manager fills in the required information and clicks "Save". - The system validates all fields, generates the promotion record, and displays the message: "Promotion created successfully".
Alternative Flow	None.
Exception Flow	3a. Missing required fields - At step 3, the system detects empty required fields and displays: "Please fill in all required fields". - Return to step 3 of the Main Flow. 4a. Duplicate promotion code - At step 4, the system detects the promotion code already exists and displays: "Promotion code already exists. Please use a different code". - Return to step 3 of the Main Flow. 4b. Invalid date range - At step 4, the system detects end date is before start date, or start date is in the past, and displays: "Invalid promotion date range". - Return to step 3 of the Main Flow.

	4c. Invalid discount value - At step 4, the system detects discount value ≤ 0 or percentage > 100 and displays: "Invalid discount value". - Return to step 3 of the Main Flow.
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Edit Promotion	
Attributes	Detailed Specification
Use Case Name	Edit Promotion
Actor	Manager
Purpose	Allows the Manager to update the details of an existing promotion.
Pre-Conditions	The Manager is logged into the system. At least one promotion exists.
Post-Conditions	The promotion is updated with the new information in the system.
Main Flow	- The Manager navigates to the Promotion Management screen and selects a promotion to edit. - The system displays the Edit Promotion form pre-filled with current promotion details. - The Manager modifies the desired fields and clicks "Save". - The system validates the updated fields, saves the changes, and displays the message: "Promotion updated successfully".
Alternative Flow	3a. Promotion is currently ACTIVE - At step 3, the system detects the promotion is currently active and restricts editing of promotion code, discount type, and discount value. - The system displays: "Some fields cannot be edited while the promotion is active". - The Manager may only edit end date, usage limit, or deactivate the promotion. - Continue to step 4 of the Main Flow.

Exception Flow	4a. Invalid date range - At step 4, the system detects end date is before start date and displays: "Invalid promotion date range". - Return to step 3 of the Main Flow. 4b. Invalid discount value - At step 4, the system detects an invalid discount value and displays: "Invalid discount value". - Return to step 3 of the Main Flow.
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Remove Promotion	
Attributes	Detailed Specification
Use Case Name	Remove Promotion
Actor	Manager
Purpose	Allows the Manager to remove a promotion that is no longer needed.
Pre-Conditions	The Manager is logged into the system. At least one promotion exists.
Post-Conditions	The promotion is removed from the system and can no longer be applied by customers.
Main Flow	- The Manager navigates to the Promotion Management screen and selects a promotion to remove. - The system displays a confirmation dialog: "Are you sure you want to remove this promotion? This action cannot be undone". - The Manager confirms the action. - The system removes the promotion record and displays the message: "Promotion removed successfully".
Alternative Flow	3a. Manager cancels the action - At step 3, the Manager clicks "Cancel". - The system closes the confirmation dialog and returns to the Promotion Management screen without making any changes.

Exception Flow	4a. Promotion is currently ACTIVE and has been used 1. At step 4, the system detects the promotion has already been applied to existing orders and displays: "Cannot remove a promotion that has been used in orders. You may deactivate it instead". 2. The system offers the option to deactivate the promotion.
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View Promotion Performance	
Attributes	Detailed Specification
Use Case Name	View Promotion Performance
Actor	Manager
Purpose	Allows the Manager to view the usage statistics and effectiveness of a promotion.
Pre-Conditions	The Manager is logged into the system. At least one promotion exists.
Post-Conditions	The system displays promotion performance data. No data is modified.
Main Flow	1. The Manager navigates to the Promotion Management screen and selects a promotion to view performance. 2. The system displays the Promotion Performance screen including: total usage count, total discount amount given, number of orders using this promotion, and revenue generated from those orders. 3. The Manager reviews the information.
Alternative Flow	None.

Exception Flow	2a. No usage data available At step 2, the system detects the promotion has not been used by any customer yet and displays: "No performance data available for this promotion".
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Create Suppliers	
Attributes	Detailed Specification
Use Case Name	Create Suppliers
Actor	Manager
Purpose	Allows the Manager to add a new supplier to the system.
Pre-Conditions	The Manager is logged into the system.
Post-Conditions	A new supplier is saved in the system and available for selection when creating Purchase Orders.
Main Flow	The Manager navigates to the Supplier Management screen and clicks "Add Supplier". The system displays the Add Supplier form including: supplier name, contact person, phone number, email, address, and tax code. The Manager fills in the required information and clicks "Save". The system validates all fields, saves the supplier record, and displays the message: "Supplier added successfully".
Alternative Flow	None.

Exception Flow	3a Missing required fields: The system detects empty required fields and displays: "Please fill in all required fields". Return to step 3. 4a Duplicate supplier name: The system detects the supplier name already exists and displays: "Supplier name already exists. Please use a different name". Return to step 3. 4b Duplicate tax code: The system detects the tax code already exists and displays: "Tax code already exists". Return to step 3. 4c Database connection error: The system cannot connect and displays: "Unable to connect to the system. Please try again later".
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Edit Suppliers	
Attributes	Detailed Specification
Use Case Name	Edit Suppliers
Actor	Manager
Purpose	Allows the Manager to update the information of an existing supplier.
Pre-Conditions	The Manager is logged into the system. At least one supplier exists.
Post-Conditions	The supplier information is updated in the system.
Main Flow	The Manager navigates to the Supplier Management screen and selects a supplier to edit. The system displays the Edit Supplier form pre-filled with current supplier information. The Manager modifies the desired fields and clicks "Save". The system validates the updated fields, saves the changes, and displays the message: "Supplier updated successfully".
Alternative Flow	3a Supplier has active Purchase Orders:

	The system detects the supplier has Purchase Orders with status PENDING or CONFIRMED and restricts editing of supplier name and tax code. The system displays: "Some fields cannot be edited while there are active Purchase Orders". The Manager may only edit contact information. Continue to step 4.
Exception Flow	4a Duplicate supplier name: The system detects the new name already belongs to another supplier and displays: "Supplier name already exists". Return to step 3. 4b Duplicate tax code: The system detects the new tax code already belongs to another supplier and displays: "Tax code already exists". Return to step 3. 4c Database connection error: The system cannot connect and displays: "Unable to connect to the system. Please try again later".

Remove Suppliers	
Attributes	Detailed Specification
Use Case Name	Remove Suppliers
Actor	Manager
Purpose	Allows the Manager to remove a supplier that is no longer in use.
Pre-Conditions	The Manager is logged into the system. At least one supplier exists.
Post-Conditions	The supplier is removed from the system and can no longer be selected in Purchase Orders.
Main Flow	The Manager navigates to the Supplier Management screen and selects a supplier to remove. The system displays a confirmation dialog: "Are you sure you want to remove this supplier? This action cannot be undone". The Manager confirms the action. The system removes the supplier record and displays the message: "Supplier removed successfully".

Alternative Flow	3a Manager cancels the action: The Manager clicks "Cancel". The system closes the dialog and returns to the Supplier Management screen without making any changes.
Exception Flow	4a Supplier has active Purchase Orders: The system detects the supplier has Purchase Orders with status PENDING, CONFIRMED, or SHIPPING and displays: "Cannot remove supplier with active Purchase Orders. Please cancel or complete all related orders first". 4b Database connection error: The system cannot connect and displays: "Unable to connect to the system. Please try again later".

Create Purchase Order	
Attributes	Detailed Specification
Use Case Name	Create Purchase Order
Actor	Manager, Staff
Purpose	Allows the Manager or Staff to create a purchase order to import products from a supplier.
Pre-Conditions	The actor is logged into the system. At least one supplier exists. At least one product exists in the catalog.
Post-Conditions	A new Purchase Order is saved in the system with status PENDING.
Main Flow	The actor navigates to the Purchase Order screen and clicks "Create Purchase Order". The system displays the Create Purchase Order form. The actor selects a supplier from the supplier list. The actor adds products to the order by selecting product, quantity, and unit price for each item. The actor sets the expected delivery date and clicks "Submit". The system validates all fields, generates the Purchase Order with status PENDING, and displays the message: "Purchase Order created successfully".

Alternative Flow	None.
Exception Flow	5a No product added: The system detects the order has no product items and displays: "Please add at least one product to the order". Return to step 4. 6a Invalid quantity or price: The system detects quantity ≤ 0 or unit price ≤ 0 and displays: "Quantity and unit price must be greater than zero". Return to step 4. 6b Invalid delivery date: The system detects the expected delivery date is in the past and displays: "Expected delivery date must be in the future". Return to step 5. 6c Missing required fields: The system detects empty required fields and displays: "Please fill in all required fields". Return to step 3. 6d Database connection error: The system cannot connect and displays: "Unable to connect to the system. Please try again later".

Track Purchase Order Status	
Attributes	Detailed Specification
Use Case Name	Track Purchase Order Status
Actor	Manager, Staff
Purpose	Allows the Manager or Staff to view and update the current status of a Purchase Order.
Pre-Conditions	The actor is logged into the system. At least one Purchase Order exists.
Post-Conditions	The Purchase Order status is updated. If status is set to DELIVERED, the system triggers the Import Products process.
Main Flow	The actor navigates to the Purchase Order screen and selects a Purchase Order to track. The system displays the Purchase Order details including: supplier name, product list, quantities, total value, expected delivery date, and current status. The actor updates the status to the next stage and clicks "Update".

	The system saves the new status and displays the message: "Purchase Order status updated successfully". If the new status is DELIVERED, the system automatically triggers the Import Products process to update warehouse inventory.
Alternative Flow	3a Actor only views without updating: The actor reviews the information and does not change the status. The system remains unchanged.
Exception Flow	4a Invalid status transition: The system detects the requested status transition is not allowed (e.g., PENDING → DELIVERED directly) and displays: "Invalid status transition. Please follow the correct order: PENDING → CONFIRMED → SHIPPING → DELIVERED". 4b Purchase Order already CANCELLED or DELIVERED: The system detects the order is in a final state and displays: "This Purchase Order has already been completed or cancelled and cannot be updated". 4c Database connection error: The system cannot connect and displays: "Unable to connect to the system. Please try again later".